Paritosh Marathe

917-733-1902 | pmparitoshmarathe@gmail.com | linkedin.com/in/paritosh-marathe

SUMMARY

AI Systems Engineer building computer vision, telemetry, and analytics platforms for electrical utilities and manufacturing. Led SSAM rearchitecture and deployments across 20+ field units. Experienced in edge AI, industrial protocols, secure device provisioning, and offline deployments.

EDUCATION

State University of New York, at Binghamton — Binghamton, NY
Master of Science, Computer Science, GPA: 3.51/4.00 — *August 2022 - May 2024*

Savitribai Phule Pune University — Pune, India
Bachelor of Engineering in Computer, GPA: 7.81/10.00 — *Aug 2016 – May 2020*

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, Dart, Bash, SQL
Frameworks & Libraries: FastAPI, Django, Next.js, Flutter, Pandas, NumPy
Computer Vision & Edge AI: DINOv2, DINOv3, OpenCV, PyTorch, TensorFlow, ONNX, TensorRT, NVIDIA Jetson Orin NX
AI Methods & Retrieval: Few-shot/zero-shot detection, segmentation, model distillation, RAG, GPT4All, BM25, Qdrant
Databases & Storage: PostgreSQL, TimescaleDB, MongoDB, Redis, AWS S3
Industrial & Messaging: MQTT, EMQX, DNP3, Modbus TCP, Sparkplug, UNS, RabbitMQ, Kafka, REST, SOAP
Security & Identity: Keycloak, SSO, RBAC, mutual TLS
Streaming & Embedded: FFmpeg, GStreamer, MediaMTX, WebRTC, WHEP, STUN/TURN, Raspberry Pi
Cloud & Infrastructure: AWS (SageMaker, Lambda), Azure, Docker, Kubernetes
DevOps & Testing: Git, Jenkins, Azure Pipelines, Nexus, SonarQube, Pytest, Mockito, Selenium, Postman
Monitoring & Automation: Prometheus, Grafana, Odoo
Engineering Practices: CI/CD, TDD, microservices, DAG rule engines, Agile, cloud/hybrid/offline deployments

PROFESSIONAL EXPERIENCE

SAM Analytics | *Systems Engineer* | Durham, North Carolina — January 2025 – Present

- Owned the end-to-end rewrite and rearchitecture of SSAM, moving from single-threaded scripts to multithreaded and multiprocess execution, PID-level process management, process telemetry, and packaged Windows executables.
- Rebuilt streaming with FFmpeg, MediaMTX, and WHEP. For the same five-stream workload, observed RAM fall from approximately 1.5–2 GB per 720p stream to approximately 1 GB total, and Intel i5 CPU utilization from approximately 100% to 5%; these are approximate observations, not a standardized benchmark.
- Integrated DNP3 and Modbus TCP into SSAM to deliver real-time telemetry and alarm backhaul to utility SCADA systems.
- Designed computer vision pipelines for thermographic inspections to identify overheating, degradation, and fault patterns in substations and field equipment.
- Owned deployment, field troubleshooting, and maintenance for SSAM/LADS across 20+ field units, working directly with utility clients to resolve operational issues.
- Centralized SSAM identity management with Keycloak, enabling single sign-on and federated authentication.
- Built a utility analytics platform with Next.js, Python, and TimescaleDB; engineered a directed acyclic graph (DAG) rule engine with strict governance, rule-definition tracking, and execution tracking.
- Integrated microcontroller-based edge hardware with MQTT telemetry and server-generated QR codes to support real-time monitoring and access workflows.
- Designing a kiosk provisioning workflow using mutual TLS, zero-trust access controls, and app-based QR code scanning.
- Customized Odoo interfaces and workflows for kiosk operations and inventory management.
- Developed edge AI detection pipelines with FFmpeg, ONNX, and TensorRT for deployment on NVIDIA Jetson Orin NX.
- Implemented video streaming and relay workflows with FFmpeg and MediaMTX, using WHEP for WebRTC playback and STUN/TURN for NAT traversal and relay connectivity.
- Integrated GStreamer, Raspberry Pi devices, and embedded sensors for cabinet monitoring and Peltier cooling control.
- Built an Ebox operations server with FastAPI and EMQX-based MQTT telemetry; added Prometheus and Grafana for monitoring and troubleshooting.
- Developed an SSAM bridge for Unified Namespace (UNS) communication with Sparkplug integration.
- Developed cloud, hybrid, and fully offline deployment pipelines to support SSAM/LADS in connectivity-constrained field environments.

Wolfspeed | *Solutions Architect Co-op* | Durham, North Carolina August 2023 – May 2024

- Created an MES data dictionary covering 300+ database objects to strengthen data quality and compliance documentation.
- Optimized critical data archiving workflows, contributing to a 20% improvement in data flow efficiency.
- Evaluated predictive maintenance, message queues, and MQTT-based Unified Namespace architectures for Industry 4.0 applications.
- Built a real-time Android OCR application with Flutter and Java, doubling batch-tagging efficiency.
- Developed Django APIs backed by SQL with role-based access controls to support secure, compliance-focused data access.
- Prototyped a retrieval-augmented generation (RAG) pipeline with GPT4All, combining BM25 keyword search and Qdrant vector retrieval to ground generated responses in retrieved context.

Wolfspeed | *Manufacturing Executive Systems Intern* | Durham, North Carolina May 2023 – August 2023

- Standardized Azure CI/CD pipelines for .NET, Python, and Java applications, reducing deployment cycle time by 30%.
- Integrated Azure pipelines with on-premises Nexus repositories to secure build artifact storage and access.
- Integrated SonarQube static analysis into build pipelines, reducing critical vulnerabilities by 12%.
- Automated test-driven development workflows with Pytest and Mockito, achieving 82% test coverage and supporting governance compliance.
- Developed a Python wafer-defect segmentation model using watershed and image processing, improving defect detection accuracy by 30%.

ASN Computer Education | *Instructor* **|** Pune, India August 2021 - July 2022

- Delivered 120+ hours of cloud computing and cybersecurity instruction to 50+ students.
- Taught Java and object-oriented programming to 40+ students, with 90% successfully developing functional applications.

Anveshak Technologies and Knowledge Solutions | *Associate Developer* | Pune, India September 2020 – June 2021

- Built a PDF ETL pipeline with preprocessing and data cleaning to extract contract tender information, tripling retrieval speed versus manual methods.
- Implemented a Python inference pipeline using AWS SageMaker, S3, and Lambda to process data and run model predictions.
- Automated mesh generation and point cloud extraction from depth images using the Blender API.

PROJECTS

Offline Computer Vision and Edge Deployment

- Built a local industrial-component detection pipeline with a LoRA-adapted DINOv2 teacher and DETR detection head; integrated MobileSAM/FastSAM for mask generation and proposal refinement.
- Implemented teacher-student distillation into ConvNeXt-Tiny with lightweight DETR, using Hungarian query matching, KL-divergence classification loss, and L1/GIoU box losses; added ONNX export for edge deployment.

Analyzing Topics, Framing, and Hate Speech in Online Political Engagement

- Built and deployed an ETL pipeline to ingest, process, and store 200+ GB of Reddit and 4chan API data in PostgreSQL for large-scale analysis.
- Developed an NLTK-based NLP pipeline for hate speech detection and sentiment analysis across 10,000+ user comments.
- Built Grafana dashboards to explore discussion trends, user engagement, and hate speech prevalence, supporting real-time monitoring and exploratory analysis.